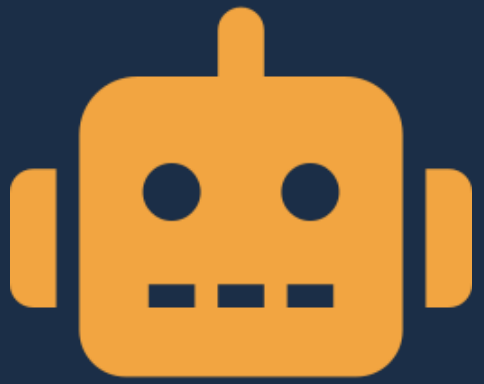
Design and Development of an Autonomous Restaurant Service Robot for Order Delivery and Customer Interaction



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Project Framework

Why restaurants need service robots

Labour shortages

Restaurants struggle to hire and keep waiting staff

Contactless service

Demand for less physical contact since COVID-19

Cheaper hardware

LiDAR, boards and sensors are now affordable

Smarter robots

Large language models enable flexible interaction

Three problems one robot must solve

1. Motion

Navigate safely and accurately between tables, chairs and moving people

2. Service

Fit the real workflow: order → kitchen → table → pickup confirmation

3. Interaction

Handle changing orders and recover from mistakes without burdening customers

Our framework: one low-cost robot that integrates all three layers

Literature Review 1/3: Map-Based Autonomous Navigation

Motion layer

Autonomous navigation of a low-cost service robot in a dynamic restaurant environment using enhanced SLAM and GA-optimized fractional PI control

Mac et al. · Results in Engineering (Elsevier) · 2025

Approach

Visual SLAM
RTAB-Map with RGB-D camera, fused with IMU through an EKF

Hybrid path planning
A* for the global path, DWA for moving obstacles

Speed control
Fractional-order PI controller tuned by a Genetic Algorithm

Low-cost platform
Raspberry Pi 4 + Arduino running ROS

0.58 → 0.26 s

motor settling time

Rise time 0.35 → 0.17 s

Strengths

- Free navigation with no floor lines
- Repeated tests reported as mean ± SD

Limitations

- No ordering system or customer interaction
- Abstract claims contradict its own tables

Literature Review 2/3: Complete Restaurant Service Cycle

Service layer

RoboServe: Design and Implementation of a Robotic Agent for Serving Food at Restaurants
Sayor et al. · ICCA 2024 (ACM) · 2024

Approach

QR ordering
Customer scans a table QR code and orders on a website

Payment and kitchen
Online payment, database and chef dashboard

Line-following delivery
ESP32 robot follows a black line with IR sensors

Smart return
Empty-tray detection sends the robot back to the kitchen

97%

delivery to the correct table

Prototype cost ≈ 5,600 BDT

Strengths

- The only paper with a full order-to-delivery cycle
- Every feature evaluated over 100 attempts

Limitations

- Line following: any layout change means redrawing
- One order per trip, 1.5 kg payload

Literature Review 3/3: LLM Planning with One-Button Recovery

Interaction layer

Real-Time Service Robot Replanning via Simple Button Interaction for Improved Task Success and User Experience

Terashima, Yano et al. · arXiv preprint · 2026

Approach

LLM planning
GPT-4o chooses the next skill step by step (SayCan-style)

One-button feedback
Customer presses "something failed"; the LLM replans

Mobile manipulator
Toyota HSR takes voice orders and serves items

Real users
Waiter tasks tested with 224 users and a questionnaire

55% → 86%

task success rate

Customer anxiety 3.1 → 2.2 / 5

Strengths

- Recovers from errors the robot cannot detect
- Only paper tested with real users

Limitations

- Preprint, not yet peer-reviewed
- Expensive robot; no statistical test

Comparative Table of Studied Papers

Criterion	Paper 1 · Mac (2025)	Paper 2 · RoboServe (2024)	Paper 3 · Terashima (2026)
Publication	Peer-reviewed journal	Peer-reviewed conference	Preprint (arXiv)
Cost	Low	Very low	Very high
Navigation	Advanced: SLAM + A* + DWA	Primitive: line following	Not described
Ordering & payment	None	Complete: QR + payment + chef	Voice ordering, no payment
Intelligence	Weak	Moderate: voice and touch	Advanced: LLM + recovery
Error handling	None	Voice alert at obstacles	Replanning after button
User testing	None	None	224 users + questionnaire
Evaluation rigour	Good: repeats ± SD	Weak: no test conditions	Moderate: no statistics
Headline result	Settling 0.58 → 0.26 s	Correct delivery 97%	Success 55% → 86%

Strengths and Weaknesses

Paper	Main strengths	Main weaknesses
Mac et al. (2025)	• Integrated ROS navigation (SLAM, EKF, A*, DWA) • GA-tuned controller, clear response gain • No floor lines needed	• No algorithmic novelty • No obstacle-detection accuracy • No ordering or interaction
RoboServe (2024)	• Full cycle from order to payment • Empty-tray auto return • Very low cost, simple algorithms	• Inflexible line following • Unsafe timed obstacle detour • One order, 1.5 kg payload
Terashima et al. (2026)	• LLM task planning (GPT-4o) • One-button error recovery • Tested with 224 users	• Not peer-reviewed • Simulated restaurant, no statistics • Expensive robot

Main Objectives

Build a low-cost autonomous robot that navigates without floor lines, joins the real restaurant workflow from order to pickup, and is evaluated with measurable results.

Navigation & Motion

- Map-based navigation, no floor lines
- Safe avoidance of moving people
- Accurate localisation (EKF)
- Smooth, spill-free motor control

Service System

- QR ordering linked to kitchen and robot
- Pickup confirmation and auto return

Interaction

- Clear status screen and voice alerts

Hardware & Safety

- Low-cost available components
- Sturdy body, ≥ 3 kg, ≥ 3 h
- E-stop and ≤ 0.5 m/s speed

Testing & Reporting

- Quantitative test of every function
- Real-user trials with statistics
- Consistent documentation

Secondary Objectives



Multi-order delivery

Carry several tables' orders per trip on separate shelves, visiting tables in the shortest order

Fixes RoboServe's one-order limit



LLM task planning on low-cost hardware

An LLM chooses from a fixed list of robot skills, with an offline fallback mode

Brings Paper 3's intelligence at low cost



One-button error correction

A "something went wrong" button triggers automatic replanning

Paper 3 raised success 55% → 86%



Research gap we address

No studied paper combines autonomous navigation, a complete service system and intelligent error recovery in one low-cost restaurant robot.

Technical Specifications (1/2)

Category	Parameter	Specification
Operating environment	**Workspace**	Indoor restaurant, flat and smooth floor, aisles ≥ 70 cm
	Conditions	Fixed or variable indoor lighting; static obstacles and moving people
Physical	**Dimensions**	Footprint ≤ 50 × 50 cm, height ≤ 120 cm; passes a 60 cm opening
	Mass	Total mass ≤ 25 kg
	Workspace	Any mapped table; tray height 75–110 cm
Payload & dynamics	**Payload**	Carry ≥ 3 kg on 3 shelves
	Dynamics	Acceleration ≤ 0.5 m/s^2 so that drinks do not spill
Actuators & power	**Speed**	Linear ≥ 0.3 m/s (capped at 0.5 m/s), angular ≥ 0.8 rad/s at full load
	Positioning accuracy	Stop at the table within ± 100 mm and ± 5°
	Power autonomy	≥ 3 h of continuous work at full load

Technical Specifications (2/2)

Category	Parameter	Specification
Sensors	**Obstacle detection**	360° LiDAR, range 0.15 m to 6 m
	Orientation	IMU heading accuracy ± 1°
Processing	**Update rate**	Navigation and obstacle avoidance ≥ 10 Hz
	Latency	Perception to action ≤ 100 ms
Communication	**Data & range**	Wi-Fi, sensor and status data ≥ 10 msg/s, stable over ≥ 30 m
	Interface	Touchscreen with status, battery and errors; manual / automatic switch
Safety & reliability	**E-Stop**	Physical and software E-Stop, stop in < 0.5 s
	Protection	Fuses for over-current; insulated, shielded wiring against EMI
	Safe mode	Stop on Wi-Fi loss or obstacle < 30 cm; return to base at battery < 15%

Thank You

References

[1] T.-T. Mac et al., "Autonomous navigation of a low-cost service robot in a dynamic restaurant environment using enhanced SLAM and GA-optimized fractional PI control," Results in Engineering, vol. 28, 108065, 2025.

[2] J. J. A.-M. Sayor et al., "RoboServe: Design and Implementation of a Robotic Agent for Serving Food at Restaurants," ICCA 2024, ACM, pp. 588–597, 2024.

[3] R. Terashima, Y. Yano, K. Arimura, H. Tamukoh, "Real-Time Service Robot Replanning via Simple Button Interaction for Improved Task Success and User Experience," arXiv:2609.17541, 2026.